

Annotation Reviewer

App Guide

Overview

The Annotation Reviewer is a browser-based tool built as part of this project to assess the quality of pixel-level segmentation masks in the Cityscapes dataset. It overlays the segmentation mask on top of the original RGB image with adjustable transparency, allowing reviewers to directly compare annotation boundaries with the actual object edges in the photograph.

The tool supports three object classes: Car, Person, and Cyclist. For classification guidelines, refer to the Annotation Quality Ontology document.

Setup

The tool requires two files from the project GitHub repository: `app.py` and `index.html`. Both files must be stored in the same folder on your machine, alongside the Cityscapes dataset with the corresponding 3-class masks.

Step 1: Download the dataset

Go to the project GitHub repository and follow the link to the Cityscapes dataset download page. Download the images and the fine annotations (gtFine).

Step 2: Generate the 3-class masks

Run the `3classes.py` script from the repository. This creates the remapped segmentation masks for the three classes used in this project: Car, Person, and Cyclist.

Step 3: Store files

Place `app.py`, `index.html`, and the dataset folder in the same directory. Update the dataset path in `app.py` to match your local folder location (see the comments in the script).

Step 4: Launch the tool

Run `app.py`. Once running, open your browser and navigate to the local address shown in the terminal. The Annotation Reviewer will load.

Interface Overview

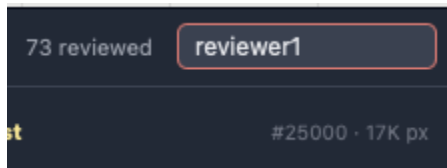


Element	Description
Image panel	Displays the current image with the mask overlay. Click on any highlighted instance to select it.
Instance list (right panel)	Lists all annotated instances in the current image by class and instance ID. Click to select an instance.
Opacity slider (top)	Controls the transparency of the mask overlay from 0% (original image only) to 100% (mask only). Use this to judge boundary alignment.
Classification form (bottom right)	Where you select the quality label, specify affected parts, and add notes before saving.
Progress bar (top)	Shows how many instances have been reviewed out of the total.
Reviewer name (top right)	Enter your name or reviewer number before starting. This is saved with every classification.

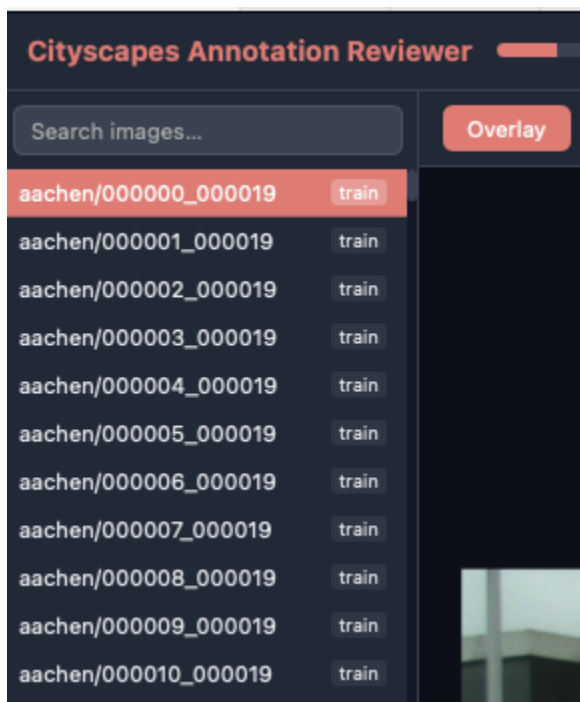
Using the Tool

Step-by-Step Workflow

1. Enter your name or reviewer number in the top right corner before starting.



2. Select the first image from the image list on the left side of the screen.

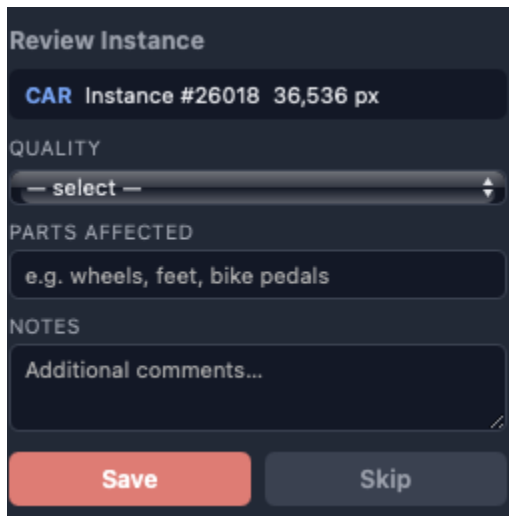


3. Use the opacity slider to adjust mask transparency. Fading the mask in and out makes it easier to see where the boundary aligns with the actual object edge.



4. Click on an instance in the instance list on the right, or click directly on a colored region in the image. The selected instance will be highlighted with a yellow border.
5. Review the mask boundary for that instance. Compare it against the actual object edge in the original photograph underneath.

6. Select the appropriate quality label from the classification form in the bottom right corner.



The image shows a 'Review Instance' form with a dark background. At the top, it says 'CAR Instance #26018 36,536 px'. Below this is a 'QUALITY' section with a dropdown menu currently showing '— select —'. Underneath is a 'PARTS AFFECTED' section with a text input field containing 'e.g. wheels, feet, bike pedals'. Below that is a 'NOTES' section with a text area containing 'Additional comments...'. At the bottom are two buttons: a red 'Save' button and a grey 'Skip' button.

7. If selecting Over-annotated, Under-annotated, or Mix, fill in the Parts Affected field describing which parts of the object are affected.
8. Add any relevant notes if needed, then press Save.
9. Move to the next instance in the list. Review all instances in the image before moving to the next image.



Opacity Adjustment

The opacity slider is the primary tool for judging annotation quality. Two reference points:

- 0% opacity: the original RGB image is fully visible with no mask. Use this to see the actual object clearly.
- 50% opacity: the mask and original image are blended equally. This is the most useful setting for comparing mask boundaries to object edges.
- 100% opacity: the mask is fully visible with no original image. Use this to inspect the mask shape in isolation.

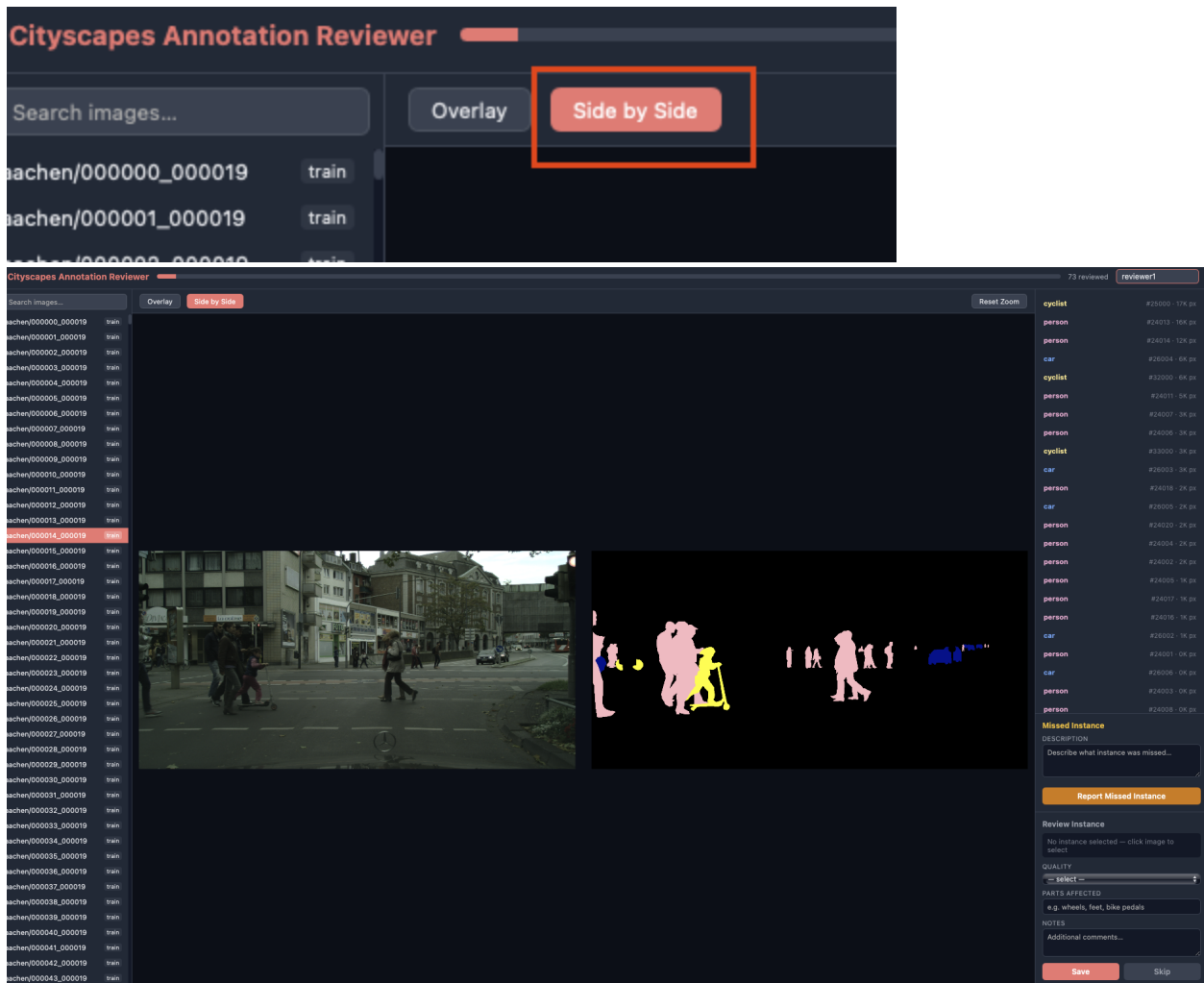
0% and 50% opacity examples



Side-by-Side Mode

Click the Side by Side button in the top right corner to switch to a split view showing the original RGB image and the colored mask next to each other. This is useful when the overlay makes it difficult to judge a specific boundary.

Click the Overlay button to return to the default overlay mode.

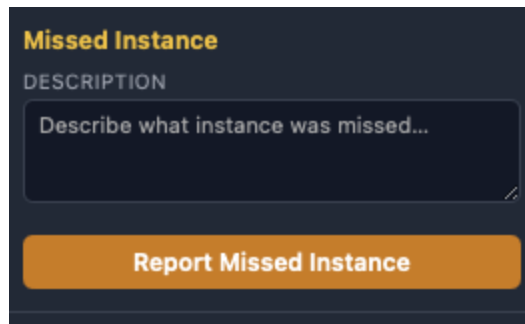


Reporting Missed Instances

If you spot an object in the image that has no annotation at all, use the Missed Instance panel on the right side of the screen. Enter a description of the object class and its approximate location in the image.

Example: "Two cars behind the fence in the upper right corner, partially visible, no mask present."

For guidance on what counts as a reportable missed instance, refer to the Missed Instances section of the Annotation Quality Ontology.

A dark-themed user interface for reporting a missed instance. It features a title 'Missed Instance' in orange, a 'DESCRIPTION' label, a text input field with the placeholder 'Describe what instance was missed...', and an orange button labeled 'Report Missed Instance' at the bottom.

Missed Instance

DESCRIPTION

Describe what instance was missed...

Report Missed Instance

Output

The tool automatically creates a CSV file in the parent folder of app.py each time you save a classification. The file contains one row per reviewed instance with the following columns:

- Image filename
- City
- Dataset split (train or val)
- Instance ID
- Class
- Quality label
- Parts affected
- Notes
- Reviewer name
- Timestamp

Do not delete or move this file during the review session. If multiple reviewers are working separately, each will produce their own CSV file which can be merged afterward.